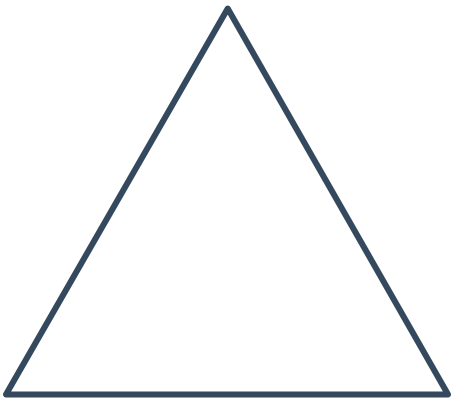
Line symmetry

1 Consider all the single digit numbers, shown below. Which of these numbers have line symmetry?

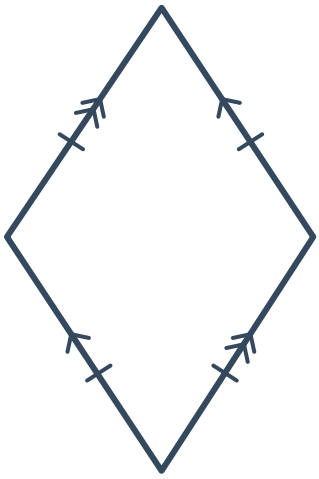
1 2 3 4 5 6 7 8 9 0

2 How many lines of symmetry do the following shapes have?

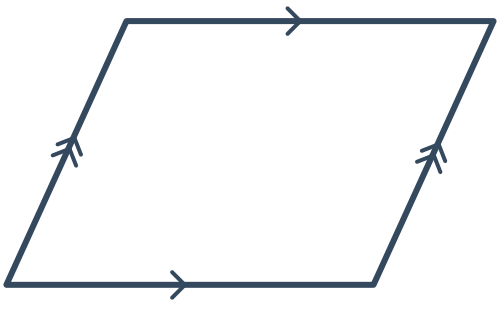
a



b



c

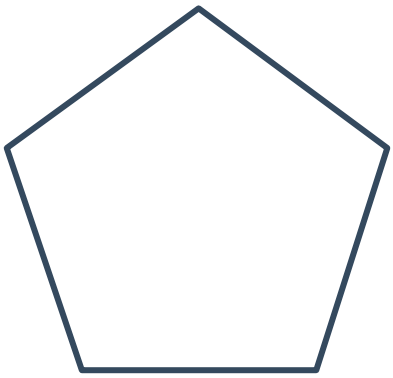


d

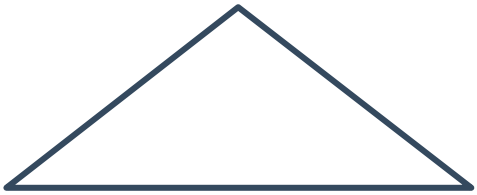


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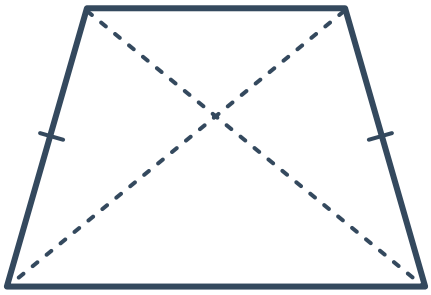
e



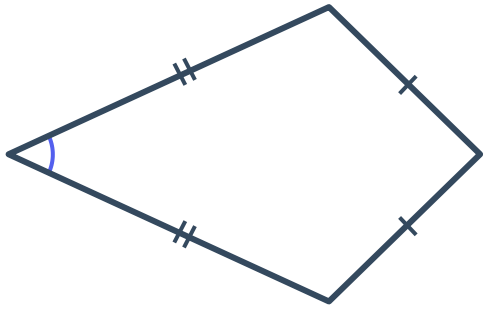
f



g



h

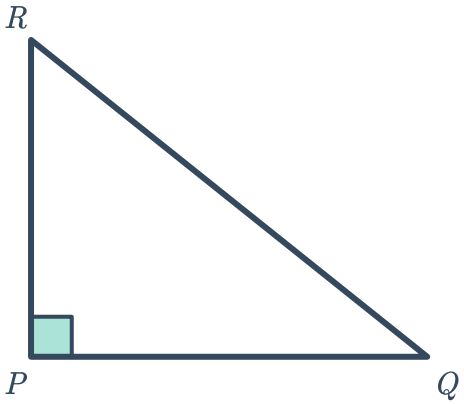


3 Consider the right-angled triangle $\triangle RPQ$ shown. The triangle is reflected across the edge PQ.

Which shape is formed by the triangle and its reflection?

A A square **B** A right-angled triangle

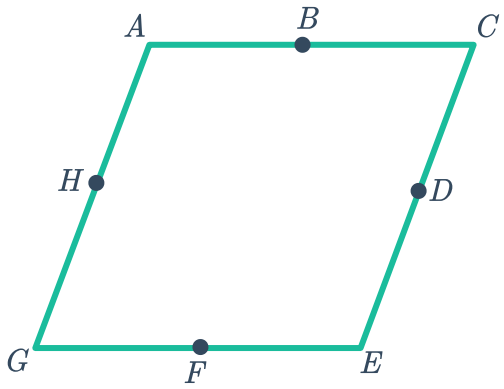
C A rectangle **D** An isosceles triangle



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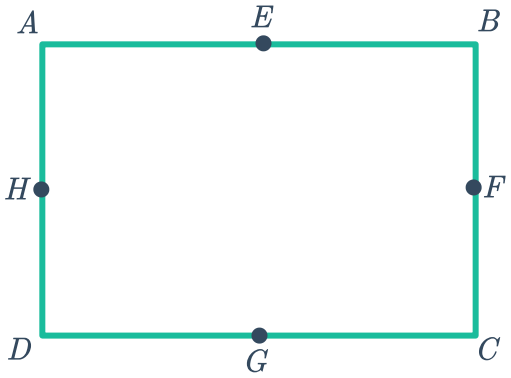
4 Consider the following rhombus. The points B, D, F and H represent the midpoints of the sides of the rhombus.

Write down the pairs of points that can be joined to form a line of symmetry for the rhombus.



5 Consider the following rectangle. The points F, G, H and E represent the midpoints of the sides of the rectangle.

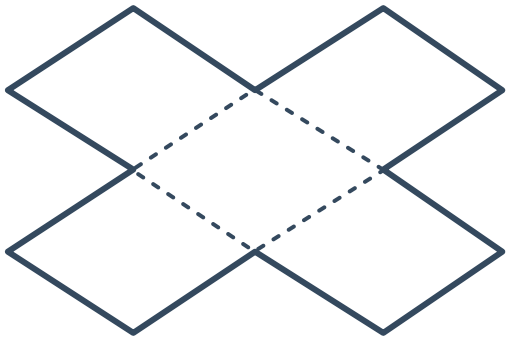
Write down the pairs of points that can be joined to form a line of symmetry for the rectangle.



6 The following shape is constructed from five identical rhombuses:

a How many lines of symmetry does this shape have?

b If each rhombus has a side length of $10\ \text{cm}$, find the perimeter P of the entire shape.

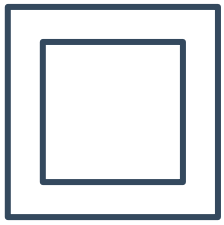


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Rotational symmetry

7 Determine whether the following shapes have rotational symmetry:

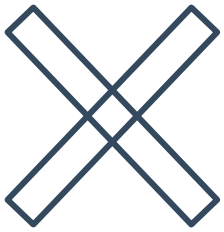
a



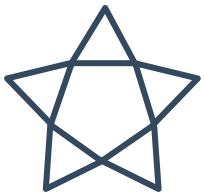
b



c

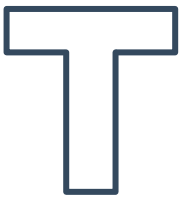


d



8 Determine whether the following figures have rotational symmetry:

a



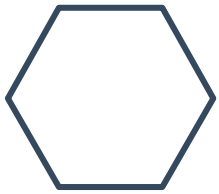
b



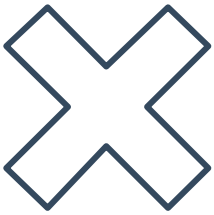
c



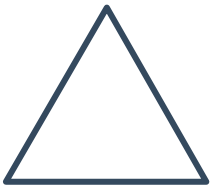
d



e



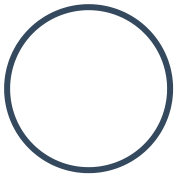
f



g



h



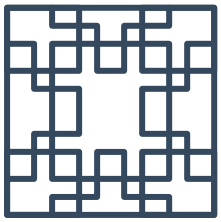
i



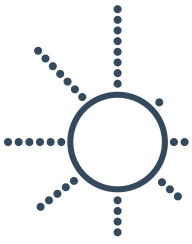
j



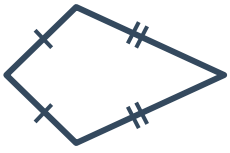
k



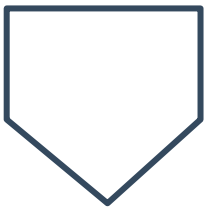
l



m



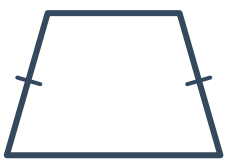
n



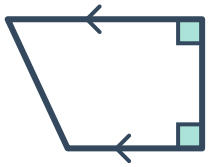
o

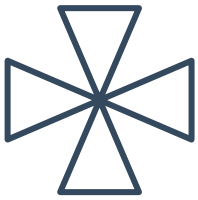


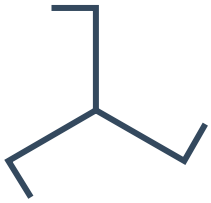
p

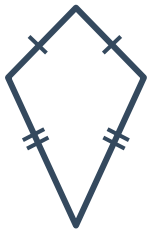


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q 

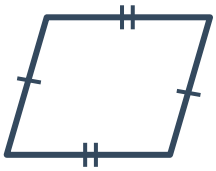
r 

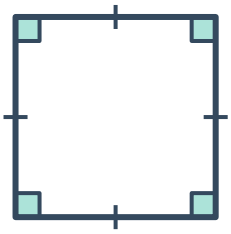
s 

t 

u 

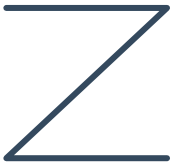
v 

w 

x 

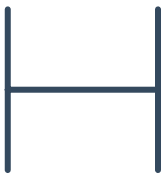
9 Determine whether the following letters have rotational symmetry:

a 

b 

c 

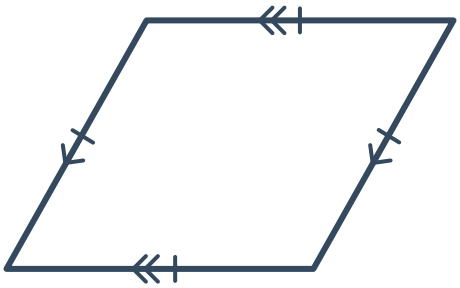
d 

e 

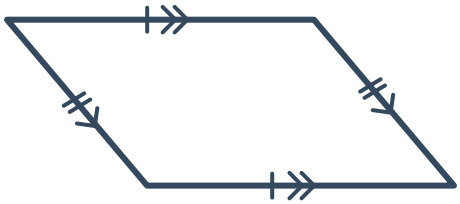
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10 State the order of rotational symmetry for the following shapes:

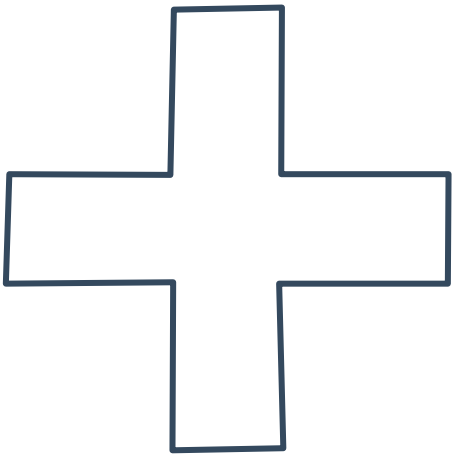
a



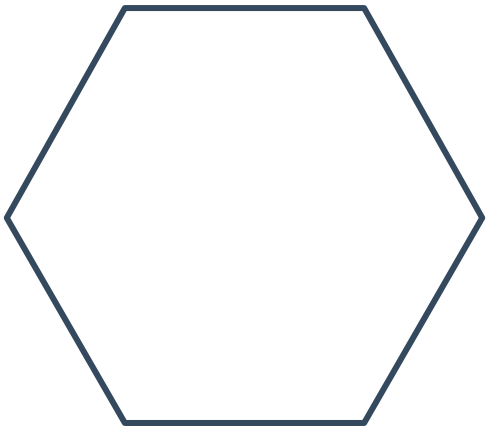
b



c



d



e

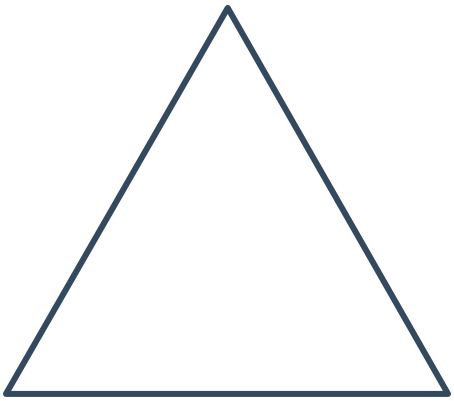


f

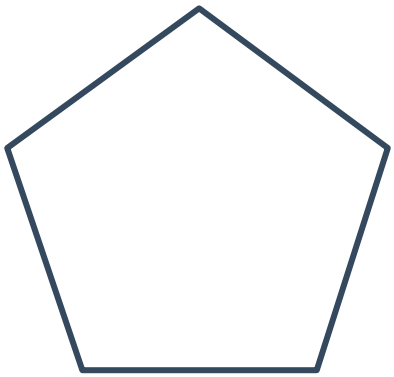


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g



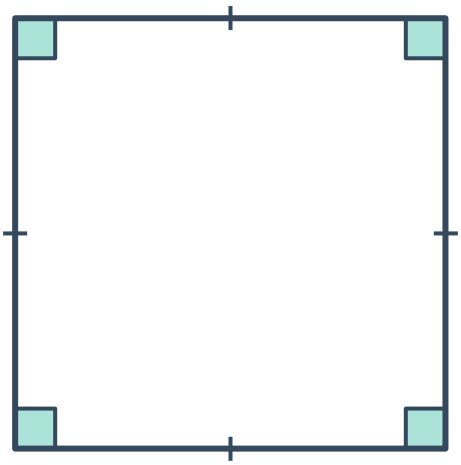
h



i



j

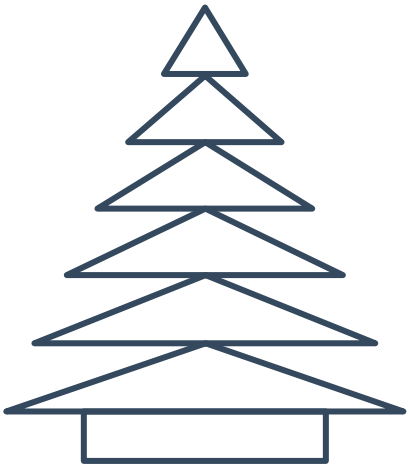


11 State the type(s) of symmetry each of the following figures have:

a

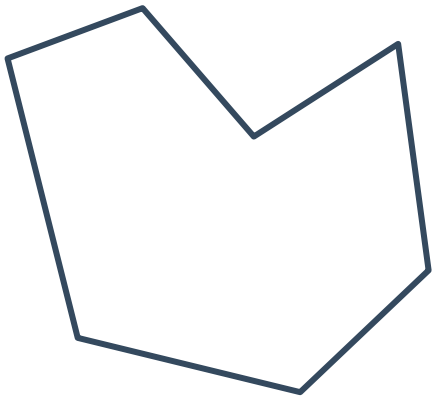


b

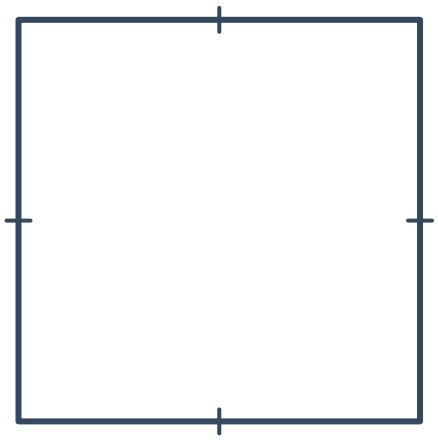


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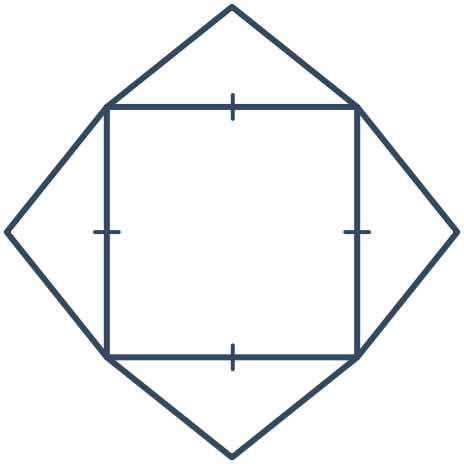
c



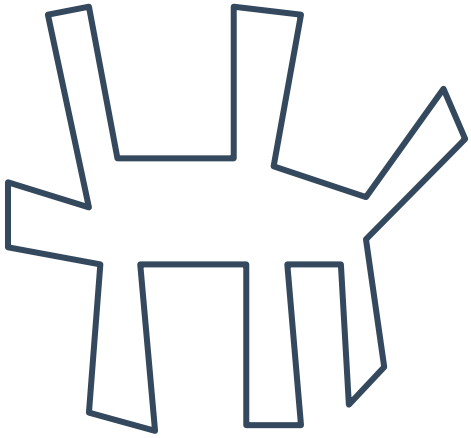
d



e



f



12 If a shape has a rotational symmetry of order 2, state the size of each angle of rotation.

13 If a shape has a rotational symmetry of order 4, state the size of each angle of rotation.

14 Do all isosceles triangles have rotational symmetry?

15 Do all equilateral triangles have rotational symmetry?

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Applications

16 Do the following road signs have rotational symmetry?

a



b



17 Consider the road sign shown:

a State the order of rotational symmetry.

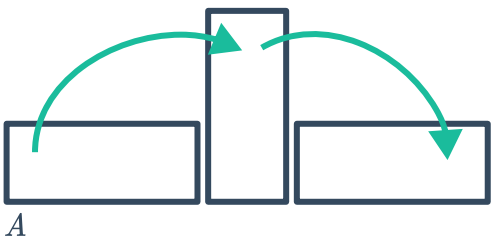
b Does the road sign have line symmetry?



18 Dave wants to get a heavy rectangular box from point A to point B by repeatedly turning it over. The width of the box is 30 cm and its length is twice its width.

a The diagram shows the distance travelled after the first and second turn. After 3 turns, what distance along the ground has been covered starting from point A?

b If the distance from point A to point B is 1620 cm, how many times must the box be turned?



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